

JMeter Performance Test Results Summary

Test Execution Summary

Total Tests Run: 16 (includes one duplicate) **Test Scenarios:** 3 (Authentication, Property Fetching, Booking Creation) **Load Levels:** 100, 200, 300, 400, 500 concurrent users

Results Overview

Authentication Tests (Login API)

Load	Samples	Avg Response Time (ms)	Min (ms)	Max (ms)	Error %	Throughput (req/sec)
100	100	209.90	196	308	0.00%	3.33
200	200	284.88	190	478	0.00%	6.67
300	300	2,152.72	247	6,277	0.00%	10.00
400	400	7,525.34	251	20,911	0.00%	13.33
500	500	14,700.21	98	35,921	1.20%	16.67

Analysis: - Excellent performance at low loads (100-200 users) - Response time increases significantly at 300+ users - Some errors at 500 users (1.2%) - Throughput scales linearly with load

Property Fetching Tests (Get Properties API)

Load	Samples	Avg Response Time (ms)	Min (ms)	Max (ms)	Error %	Throughput (req/sec)
100	500	76.31	56	226	0.00%	16.67
200	1,000	81.39	54	417	0.00%	33.33
300	1,500	1,018.29	55	4,946	0.00%	50.00
400	2,000	1,553.66	53	6,939	0.00%	66.67
500	2,500	2,314.09	55	7,945	0.00%	83.33

Analysis: - Excellent performance - fastest API - 0% errors across all load levels - Very consistent response times at low loads - Response time increases at 300+ users but remains acceptable - Highest throughput (83 req/sec at 500 users)

Booking Creation Tests (Create Booking API)

-	Load	Samples	Avg Re- sponse Time (ms)	Min (ms)	Max (ms)	Error %
100	300	135.00	59	366	12.00%	10.00
200	600	145.00	56	480	13.00%	19.80
300	900	1,480.00	56	17,988	16.56%	24.30
400	1,200	4,361.00	58	75,260	22.33%	14.70
500	1,500	6,176.00	50	59,421	27.20%	16.90

Analysis: - Error rate reduced to 12-27% after fixing token extraction and dynamic dates - Remaining errors are HTTP 400 responses: “Property is not available for the selected dates” - Root cause: only **three** seed properties are available, so concurrent 4-night bookings eventually collide even with staggered dates - Response time remains excellent for successful requests

Performance Trends

Response Time vs. Load

Authentication: - 100 users: ~210ms - 200 users: ~285ms (36% increase) - 300 users: ~2,153ms (655% increase) - 400 users: ~7,525ms (249% increase) - 500 users: ~14,700ms (95% increase)

Property Fetching: - 100 users: ~76ms - 200 users: ~81ms (7% increase) - 300 users: ~1,018ms (1,156% increase) - 400 users: ~1,554ms (53% increase) - 500 users: ~2,314ms (49% increase)

Booking Creation: - 100 users: ~135ms (when successful) - 200 users: ~145ms (7% increase) - 300 users: ~1,480ms (920% increase) - 400 users: ~4,361ms (195% increase) - 500 users: ~6,176ms (42% increase)

Key Findings

Strengths

1. **Property Fetching API** - Best performance, 0% errors
2. **Low Load Performance** - All APIs perform well at 100-200 users
3. **Throughput Scaling** - Throughput increases linearly with load

Bottlenecks Identified

1. **Authentication API** - Significant degradation at 300+ users
 - Possible causes: Database connection pool, JWT generation overhead
2. **Booking API** - Errors now stem from limited data (not auth failures)

- 12–27% of requests overlap with existing bookings because only 3 properties exist
3. **Response Time Spikes** - Max response times are very high at high loads
 - Possible causes: Resource contention, database locks, network latency

Recommendations

1. **Database Optimization**
 - Increase connection pool size
 - Add database indexes
 - Optimize queries
2. **Caching**
 - Cache property data
 - Cache authentication tokens
3. **Load Balancing**
 - Already using Kubernetes with 2 replicas
 - Consider increasing replicas at high loads
4. **Expand Booking Test Data**
 - Seed additional properties or reset bookings between runs
 - Continue using the improved Groovy processors for tokens/dates

Next Steps

1. **View HTML Reports:** Open `jmeter-results/*-report/index.html` for detailed charts
2. **Create Performance Graph:** Plot response time vs. load for all 3 scenarios
3. **Write Analysis:** Document bottlenecks and recommendations
4. **Fix Booking Tests:** Investigate and resolve 33% error rate

Files Generated

- **CSV Summary:** `jmeter-results-summary.csv`
- **HTML Reports:** `jmeter-results/*-report/index.html`
- **Raw Data:** `jmeter-results/*.jtl`